EPSR-test-ex97 Porod scattering (porodpwr, porodadd)

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises the Porod small-angle term of the `epsr` engine of EPSR [1,2]: a power-law contribution porodamp·(porodq/Q)^porodpwr [3] is added to both the data difference and the simulated fit, so that low-Q intensity from large-scale inhomogeneity is not forced into the empirical potential [4].

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (search of `nporodq`, `porodamp(in)` = `datai(nporodq,in)` + `porodadd(in)`, Porod term in the difference `dts` and in `soqism/.u01` and the R-factors)

Rust (port, rust/epsr-core/src/):

- `epsr.rs`: `Epsr::setup` (`porodpwr`, `nporodq`, `porodamp`)
- `epsr.rs`: `Epsr::iterate` (Porod term in `dts`), `write_results` (simulated fit, `.u01`, R-factors)

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [5] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [6]. For this case `porodpwr 0.5 3.5` sets porodq = 0.5 1/Å and power 3.5, and `porodadd` is 0.25, 0.10 and -0.5 for the three data sets. The amplitude of each data set is the data value at `nporodq` plus `porodadd`; a non-positive amplitude disables the term for that data set.

Inputs

Only the `.EPSR.inp` differs from EPSR-test, edited by `scripts/make_inp.sh` (via `scripts/setkw.py`): `porodpwr 0.5 3.5` and `porodadd` 0.25 / 0.10 / -0.5.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
The Q index nearest to porodq is `nporodq` = 11; the log reports Porod amplitudes 0.05630, 0.08910 and -0.36131, so data set 3 is negative and gets no Porod term, covering both branches. The Porod term changes the background-subtracted data, the fitted S(Q) (`.u01`) and the R-factors, all identical; the Porod amplitudes are recomputed from the data in every pass.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] G. Porod, "Die Röntgenkleinwinkelstreuung von dichtgepackten kolloiden Systemen I", Kolloid-Z. 124, 83-114 (1951).
[4] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[5] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[6] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[7] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.